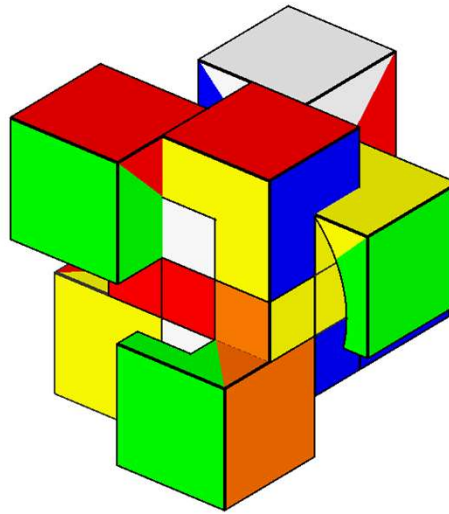


Graph Algorithms and 3D Rendering of a Complex Shapeshifting Rubik's Cube



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MNG Rämibühl 2025

Supervised by
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*Printed
2024*

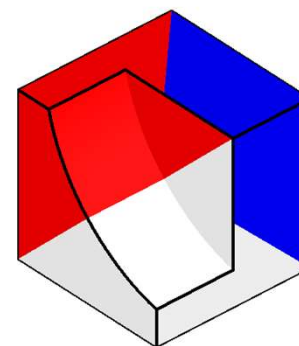
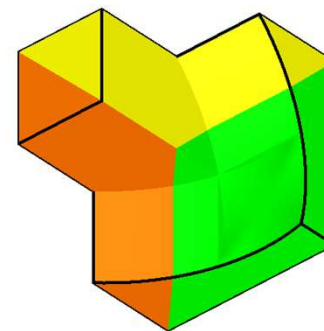
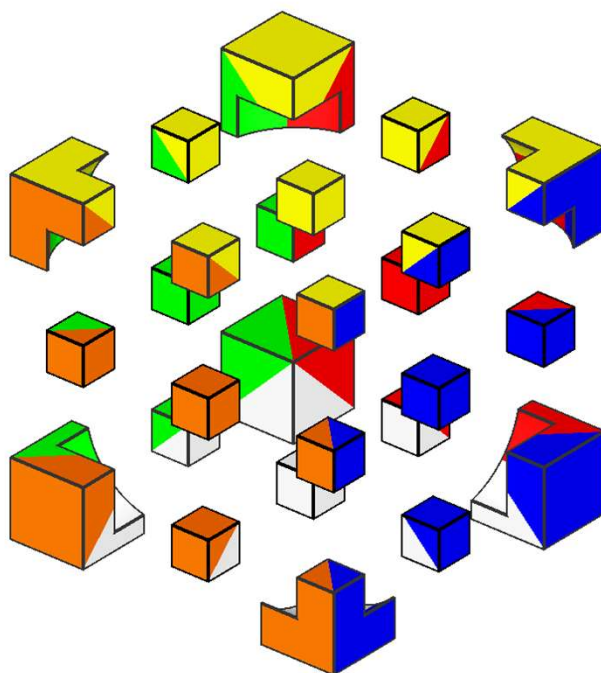
Research Statement

Which **shortest path algorithm** is capable of solving a complex, shapeshifting Rubik's cube with reasonable time and memory requirements, and how can the solution of the cube be **represented graphically** in real time?

The Puppet Cube V2

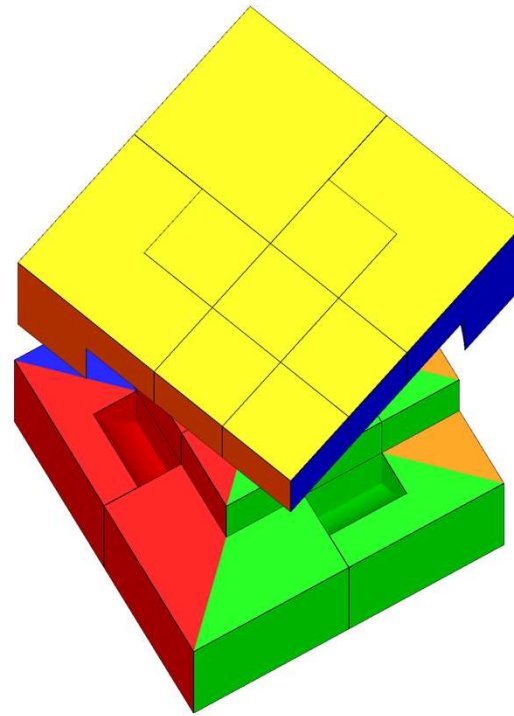
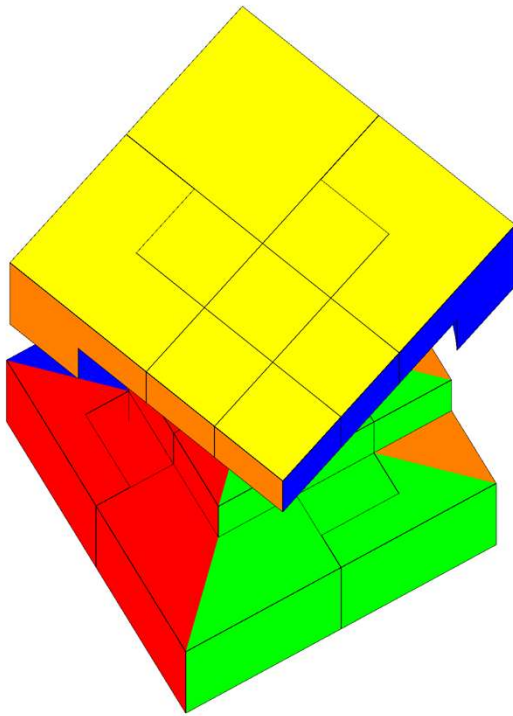
26 pieces

- 8 corners
- 12 edges
- 6 centers

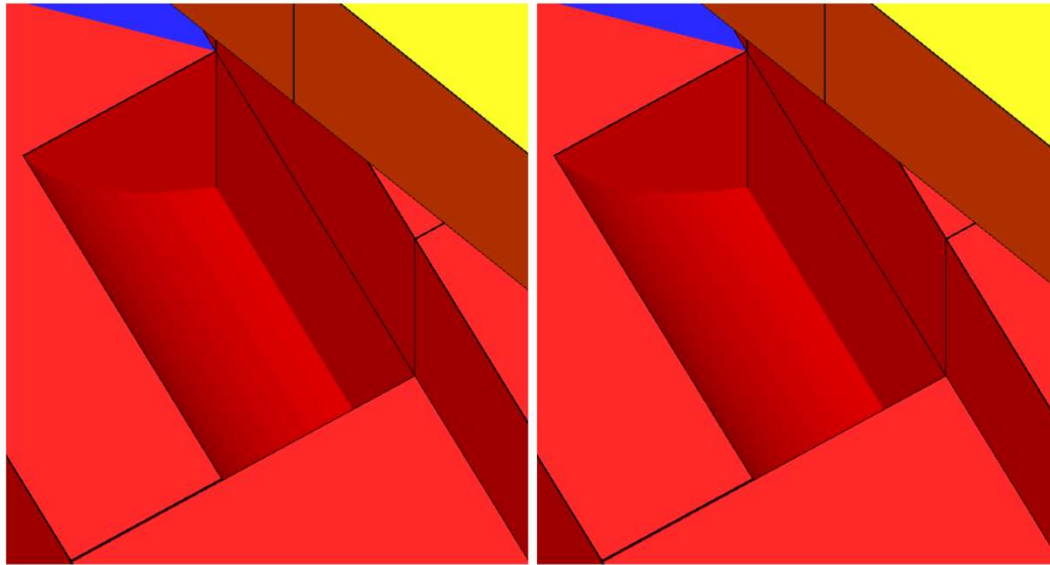


Special Corners

Visualization: Lighting

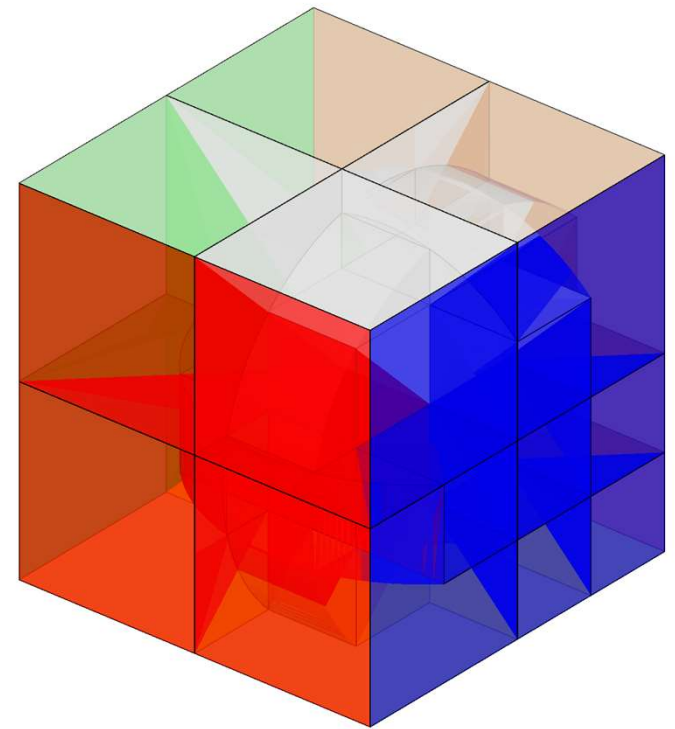


Advanced Visualization

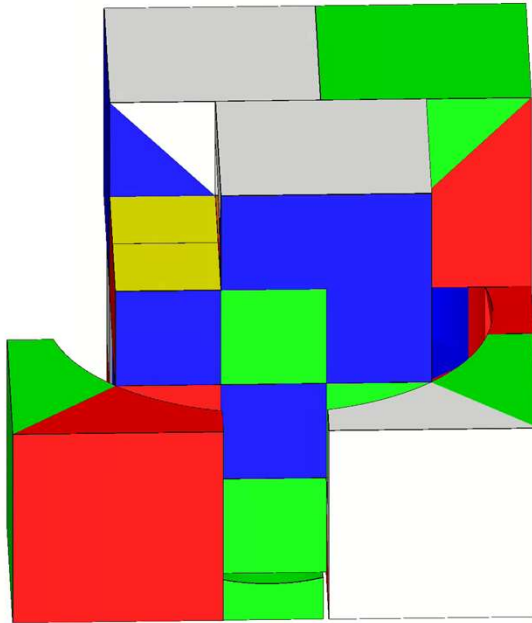


(a) Puppet Cube V2 before interpolation

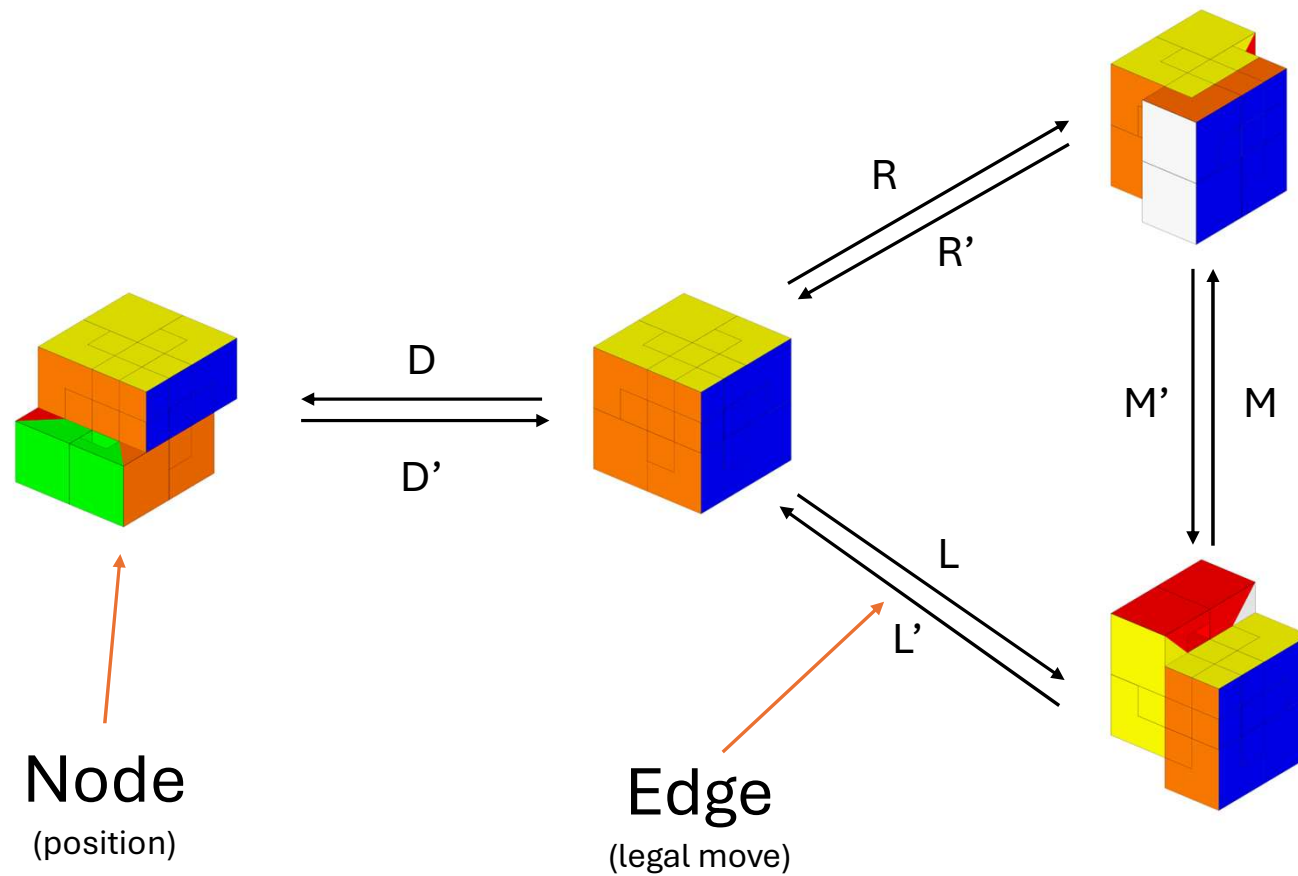
(b) Puppet Cube V2 after interpolation



In Action

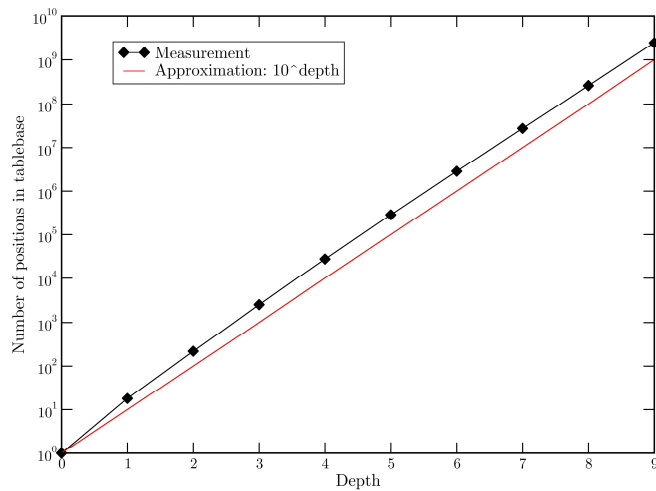


From Cubes to Graphs

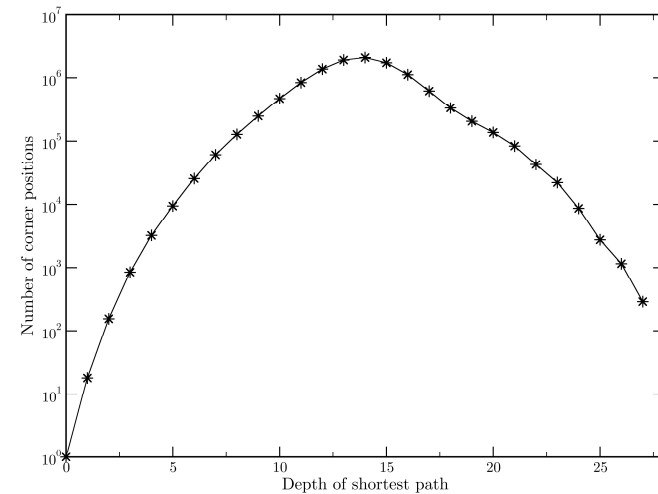


Complexity of the Graph

Full 3x3 Puppet Cube



Corners-only 2x2 Puppet Cube

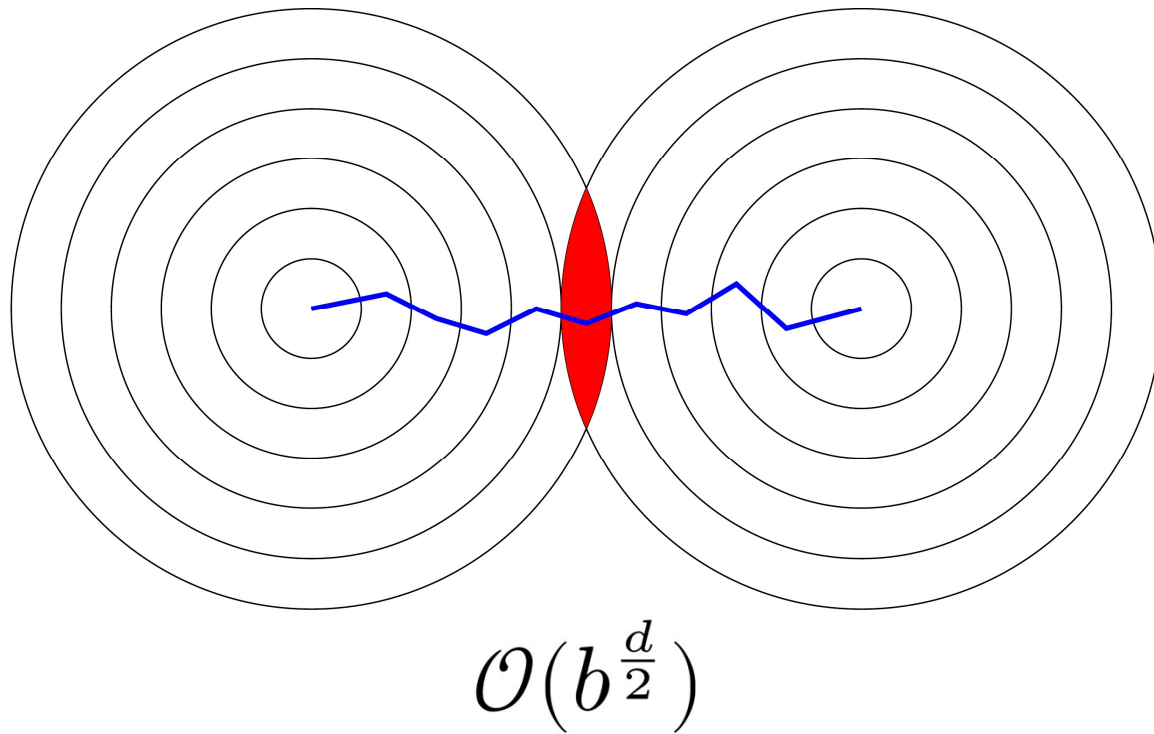


$1 \cdot 10^7$	total number of corner positions
$5 \cdot 10^{18}$	total number of positions
$3 \cdot 10^{19}$	total number of edges in the graph

5 different Algorithms

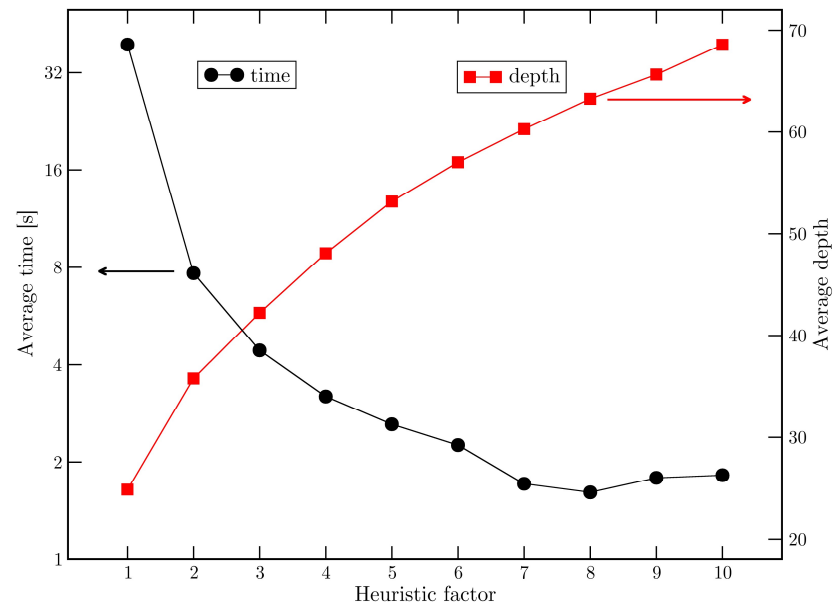
- Iterative Deepening Depth-First Search
- Visited Map / Breadth-First Search
- Meet in the Middle
- Solve Corners First
- Greedy Search using a Heuristic Function

Meet in the Middle



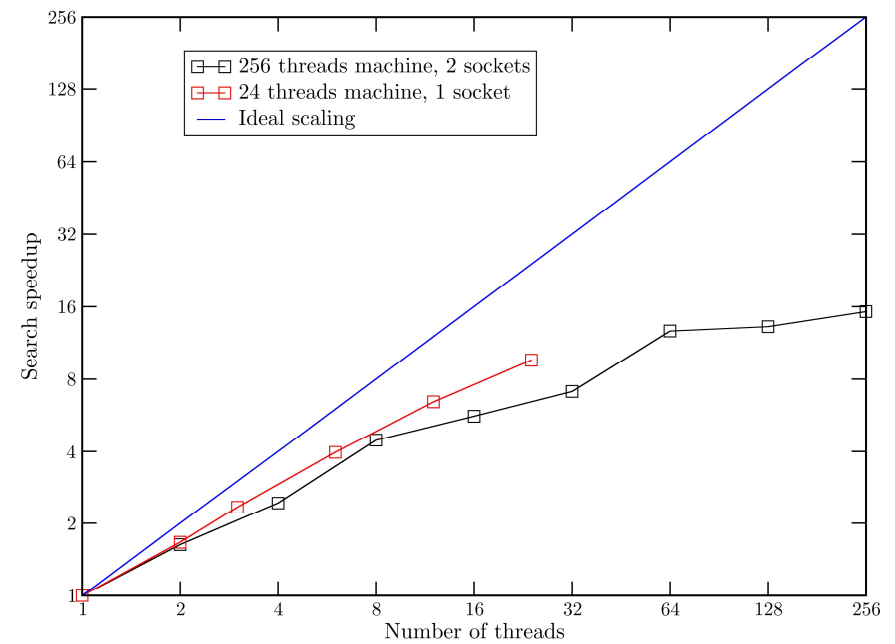
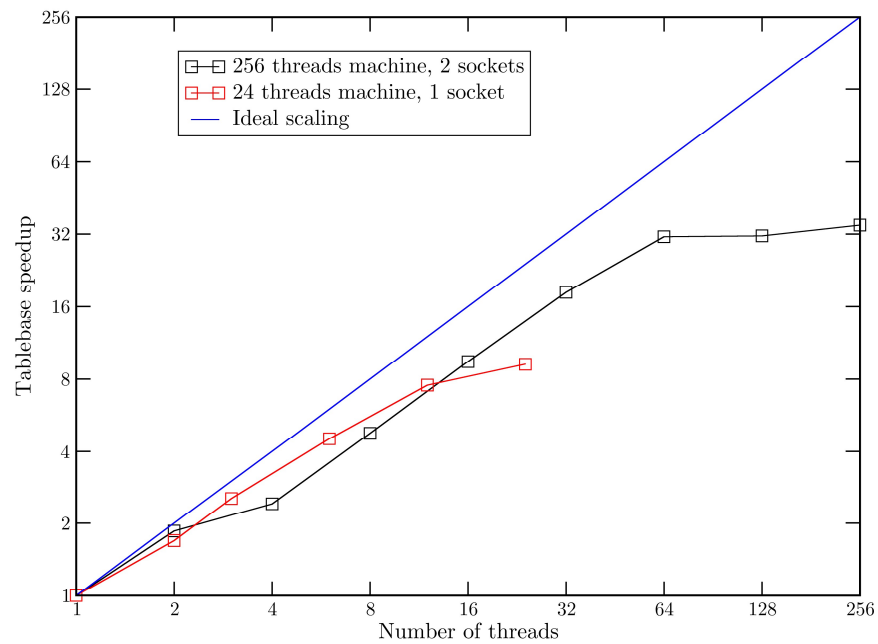
Heuristic Function:

$$h(\text{corner}, \text{edge}_1, \text{edge}_2, \text{depth}) = (\text{corner} + \text{edge}_1 + \text{edge}_2) \cdot \text{factor} + \text{depth}$$

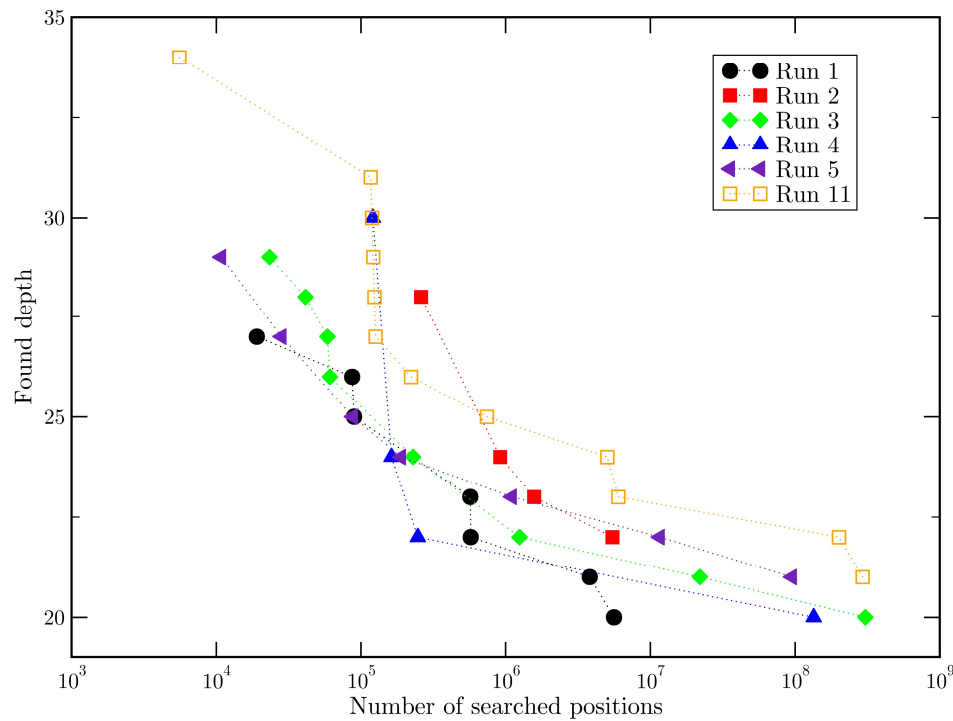


Parallelization

> 1'000'000 positions / second

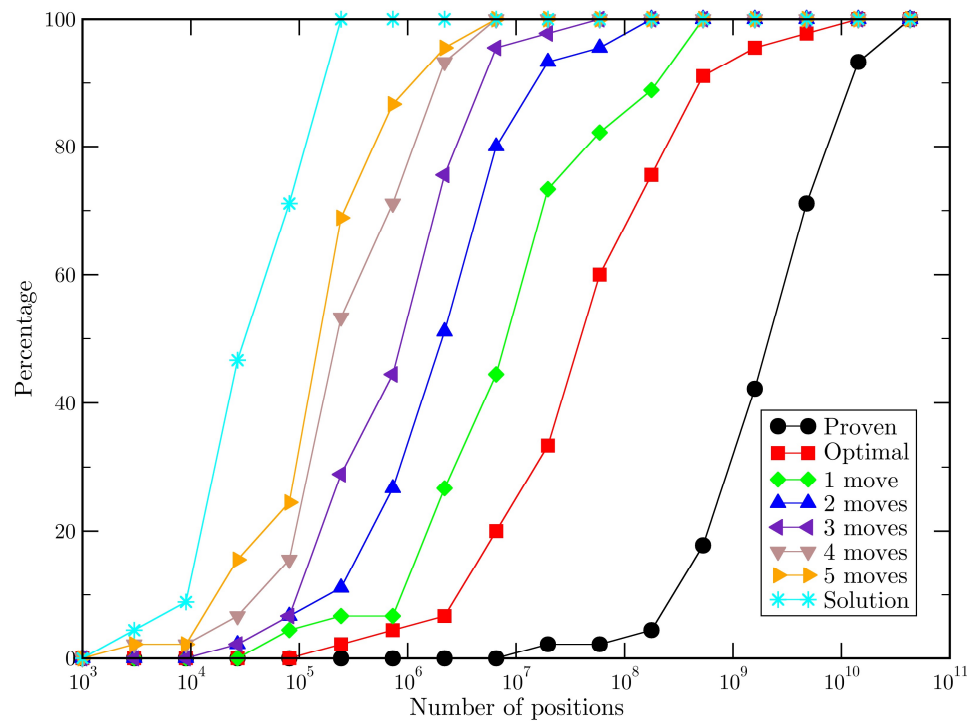


Improvement of the Found Depth



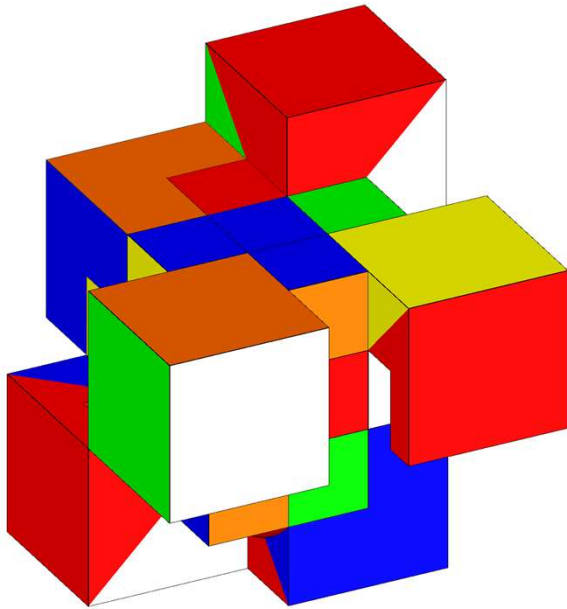
- Improves the best found depth multiple times
- Up to 11 improvements
- Average depth $\sim 20 - 21$

Quantifying Search Efficiency



- Finds solutions quickly
- Can prove optimality

Lower Bound for God's Number



Scramble:

D' L' U' L F L F D' M' D L E' F M B'
L B' U' R L B R F' F' L D D F' U' U'

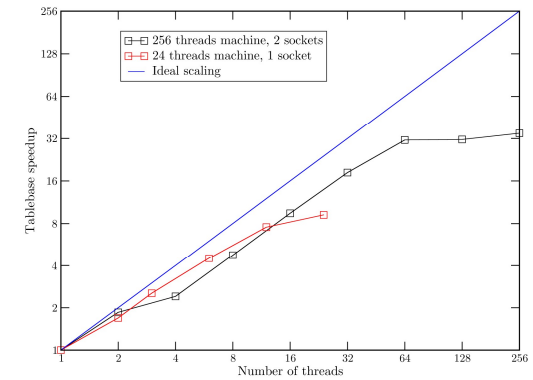
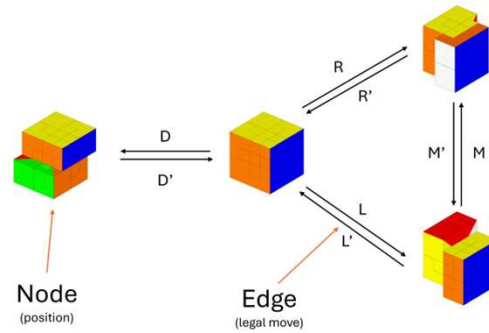
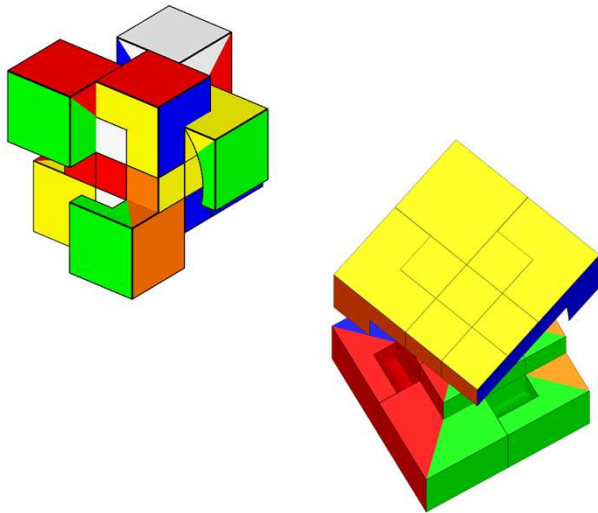
Facts:

- 30 moves needed
- $3 \cdot 10^{10}$ positions searched
- 260 GB RAM
- Solution proven to be optimal

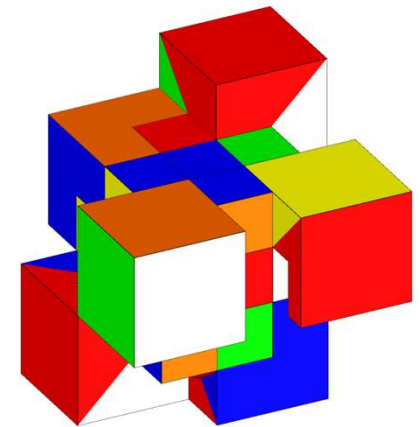
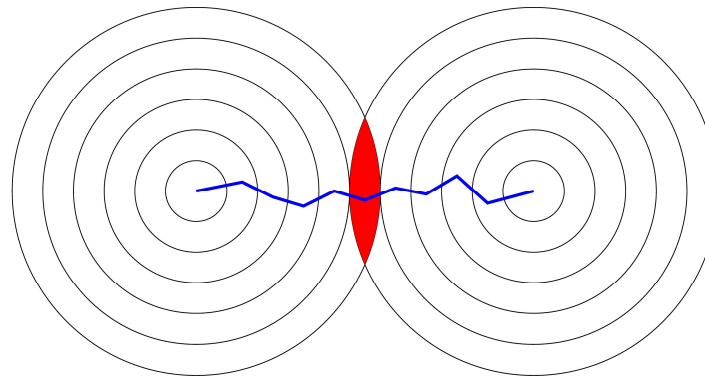
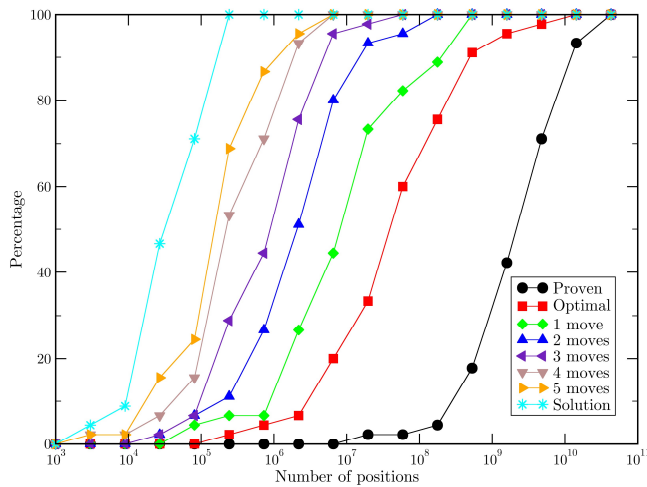
Harder than the normal Rubik's Cube!

Conclusion

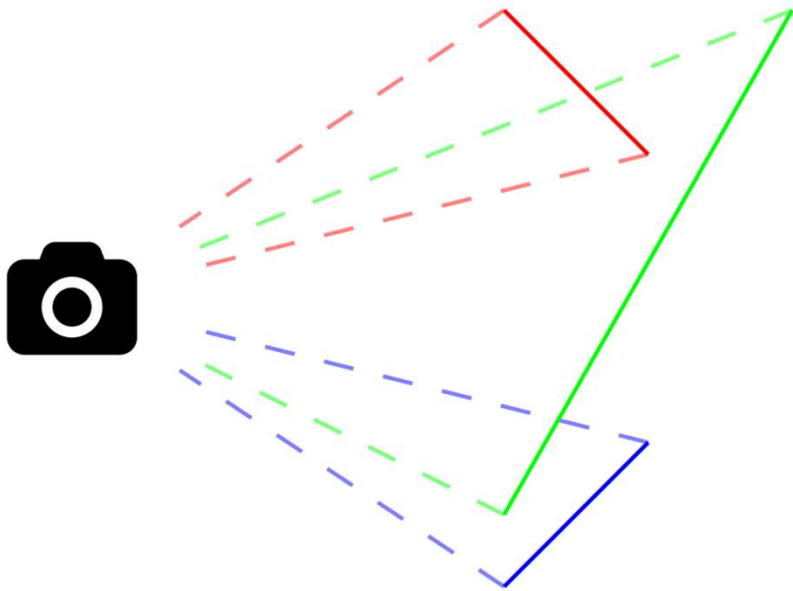
- Visualization looks like the real counterpart
 - Lighting
 - Real time
- The program can solve the Puppet Cube V2
 - Finds solutions quickly
 - Able to prove optimality



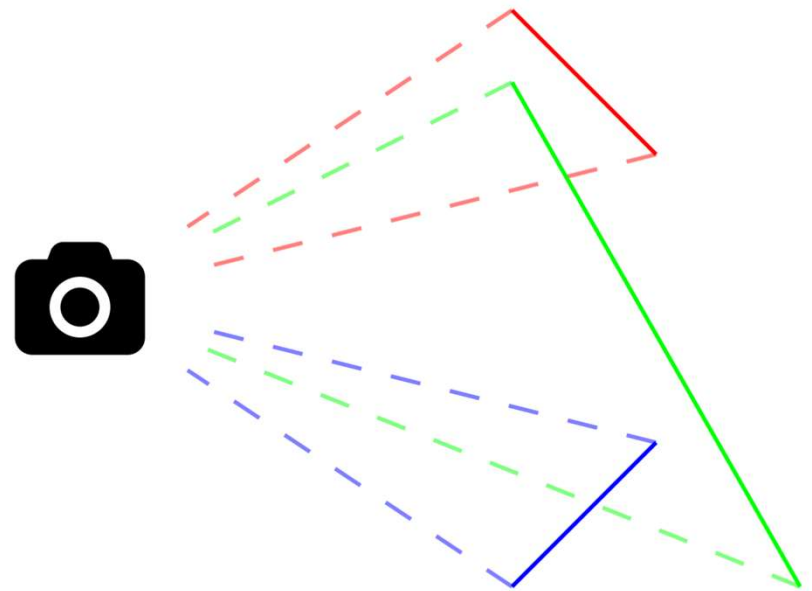
Thank you for
listening!



Transparency

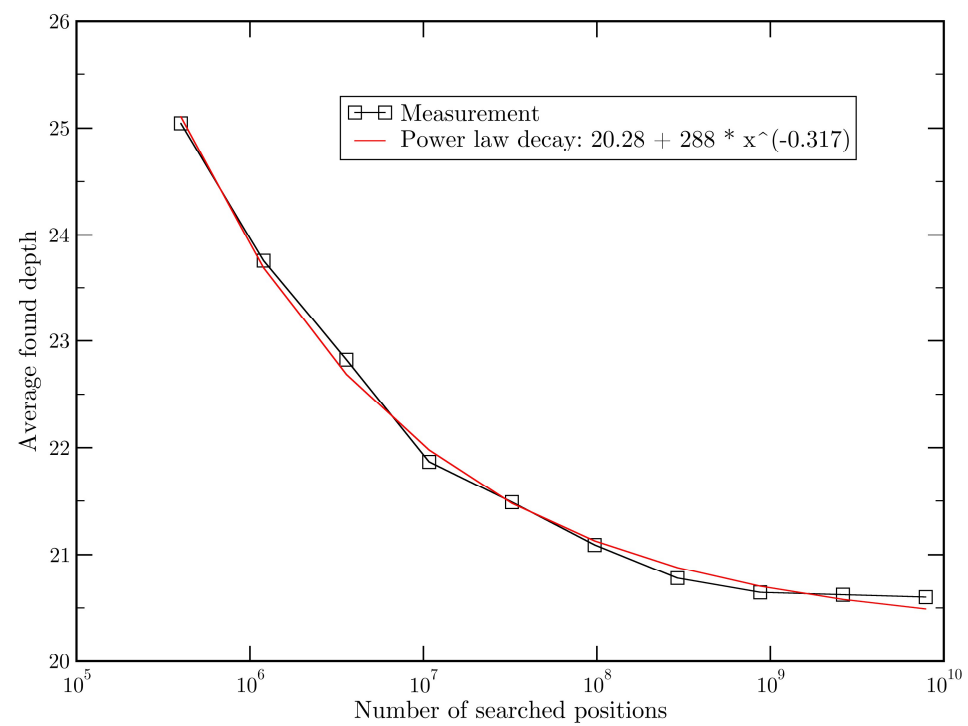


(a) Order: red, green, blue

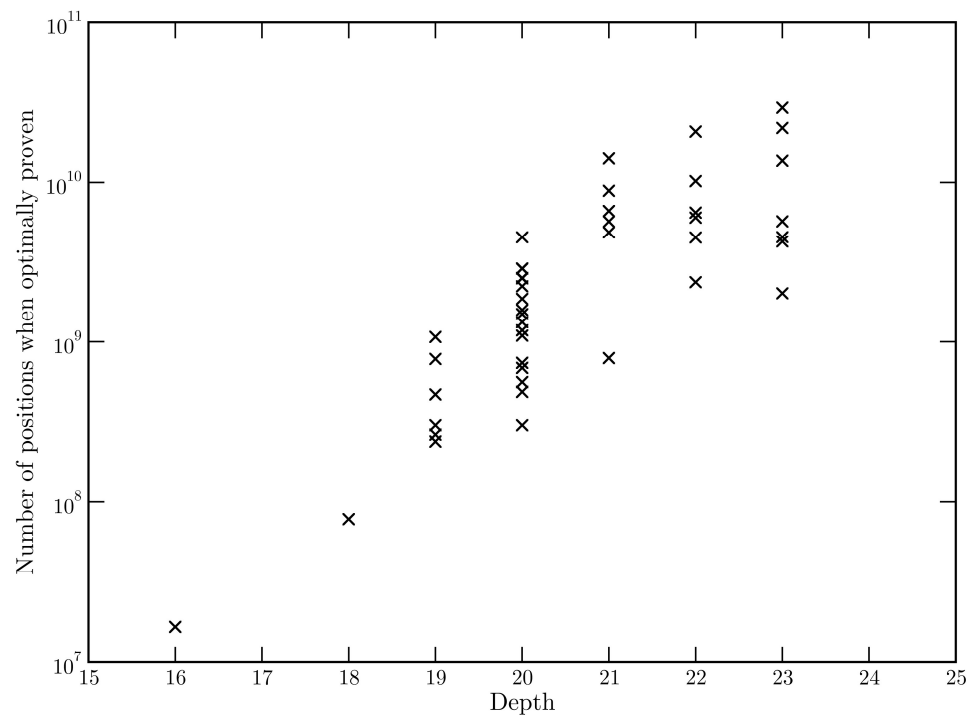


(b) Order: blue, green, red

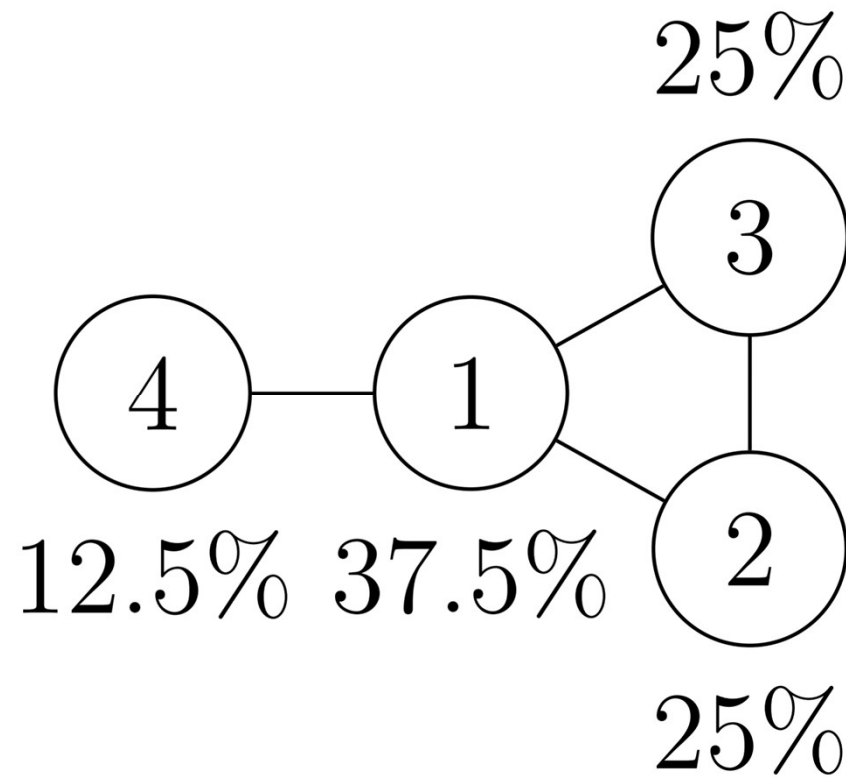
Average depth



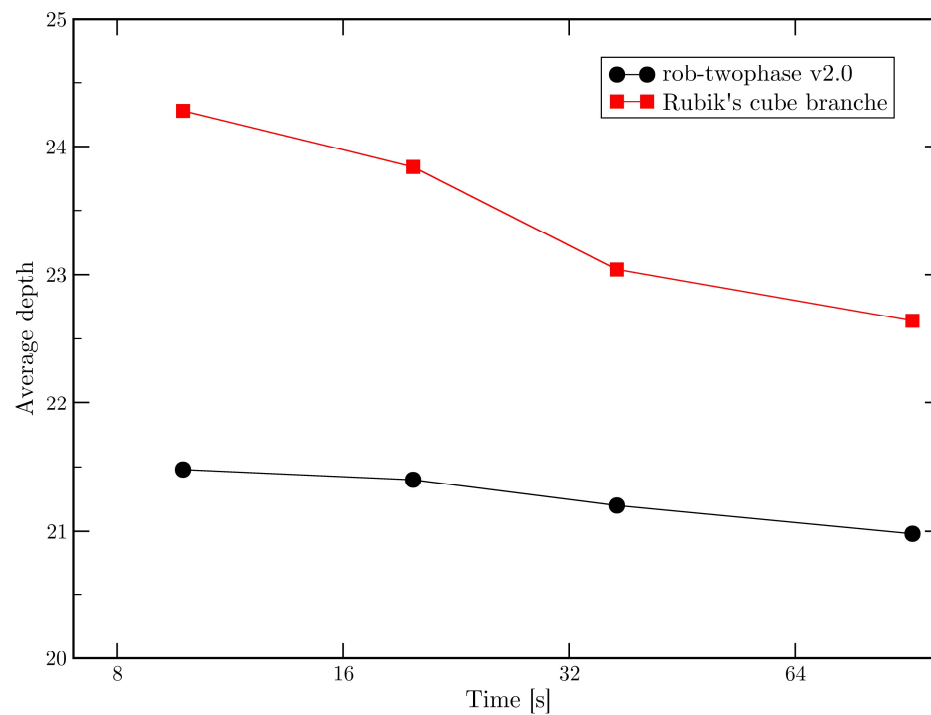
Position Proven



Random Scrambles



Comparison to one of the Best Rubik's Cube Program



- Adapted program able to solve the Rubik's Cube
- Less than 2 moves different

11,382,336	$1 \cdot 10^7$	total number of corner positions
5,583,008,927,475,302,400	$5 \cdot 10^{18}$	total number of positions
34,024,400,694,647,193,600	$3 \cdot 10^{19}$	total number of edges in the graph